

Naresh R Shanbhag

professor

基本信息

姓名: Naresh R Shanbhag

所属院校: 伊利诺伊大学厄巴纳-香槟

邮箱: shanbhag@illinois.edu

地区: United States

年龄范围: 50岁以上

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

电子与计算机工程

集成电路设计与集成系统

微电子科学与工程

研究领域

低功耗电路与能量管理

低功耗设计与能耗优化

计算机体系结构

存储器设计与架构

嵌入式系统与硬件设计

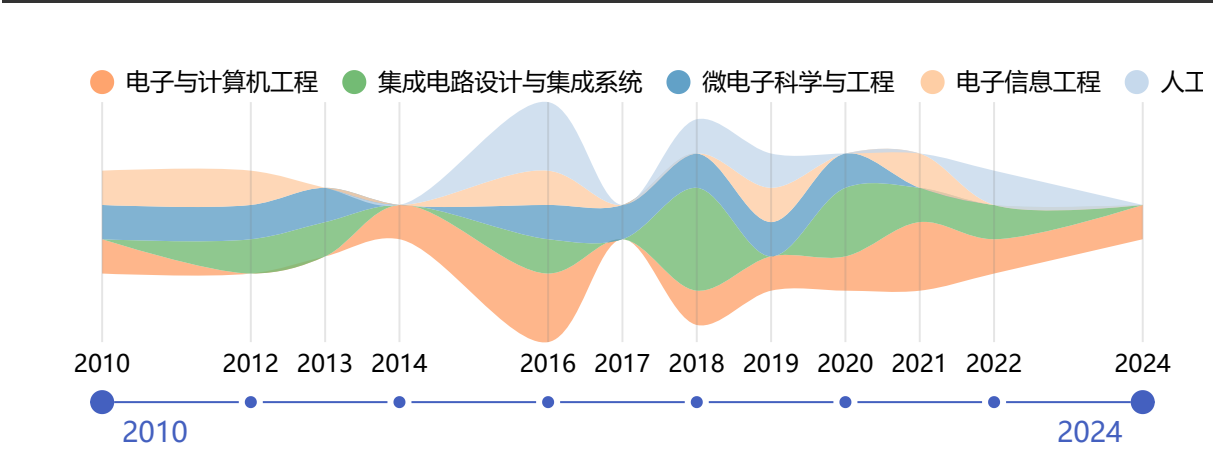
数字集成电路设计

个人简介

Naresh R Shanbhag is the Jack S. Kilby Professor of Electrical and Computer Engineering at the University of Illinois. He holds a Ph.D. in Electrical Engineering from the University of Minnesota. His teaching includes courses on digital and analog integrated circuit design, etc. His research focuses on energy - efficient machine - learning and communication systems. He has

authored books like "Deep In - memory Architectures for Machine Learning". He has received numerous honors, such as the 2024 Semiconductor Innovation Award and many IEEE - related awards. He's also been recognized as an excellent teacher at the University of Illinois.

近年发表的论文



Enhancing the Accuracy of 6T SRAM-Based In-Memory Architecture via Maximum Likelihood Detection

IEEE TRANSACTIONS ON SIGNAL PROCESSING

Kim, Hyungyo; Shanbhag, Naresh R.

被引用次数: 12024

IMPQ: REDUCED COMPLEXITY NEURAL NETWORKS VIA GRANULAR PRECISION ASSIGNMENT

2022 IEEE INTERNATIONAL CONFERENCE ON ACOUSTICS, SPEECH AND SIGNAL PROCESSING (ICASSP)

Gonugondla, Sujana Kumar; Shanbhag, Naresh R.

被引用次数: 02022

Fundamental Limits on the Computational Accuracy of Resistive Crossbar-based In-memory Architectures

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Roy, Saion K.; Patil, Ameya; Shanbhag, Naresh R.

被引用次数: 62022

Signal Processing Methods to Enhance the Energy Efficiency of In-Memory Computing Architectures

IEEE TRANSACTIONS ON SIGNAL PROCESSING

Sakr, Charbel; Shanbhag, Naresh R.

被引用次数: 102021

A 0.44-μJ/dec, 39.9-μs/dec, Recurrent Attention In-Memory Processor for Keyword Spotting

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Dbouk, Hassan; Gonugondla, Sujan K.; Sakr, Charbel; Shanbhag, Naresh R.

被引用次数: 17

2021

Deep In-Memory Architectures for Machine Learning-Accuracy Versus Efficiency Trade-Offs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kang, Mingu; Kim, Yongjune; Patil, Ameya D.; Shanbhag, Naresh R.

被引用次数: 15

2020

Deep In-Memory Architectures in SRAM: An Analog Approach to Approximate Computing

PROCEEDINGS OF THE IEEE

Kang, Mingu; Gonugondla, Sujan K.; Shanbhag, Naresh R.

被引用次数: 27

2020

An Energy-Efficient Programmable Mixed Signal Accelerator for Machine Learning Algorithms

IEEE MICRO

Kang, Mingu; Srivastava, Prakalp; Adve, Vikram; Kim, Nam Sung; Shanbhag, Naresh R.

被引用次数: 1

2019

Shannon-Inspired Statistical Computing for the Nanoscale Era

PROCEEDINGS OF THE IEEE

Shanbhag, Naresh R.; Verma, Naveen; Kim, Yongjune; Patil, Ameya D.; Varshney, Lav R.

被引用次数: 27

2019

Generalized Water-Filling for Source-Aware Energy-Efficient SRAMs

IEEE TRANSACTIONS ON COMMUNICATIONS

Kim, Yongjune; Kang, Mingu; Varshney, Lay R.; Shanbhag, Naresh R.

被引用次数: 9

2018

A 19.4-nJ/Decision, 364-K Decisions/s, In-Memory Random Forest Multi-Class Inference Accelerator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kang, Mingu; Gonugondla, Sujan K.; Lim, Sungmin; Shanbhag, Naresh R.

被引用次数: 44

2018

A Multi-Functional In-Memory Inference Processor Using a Standard 6T SRAM Array

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kang, Mingu; Gonugondla, Sujan K.; Patil, Ameya; Shanbhag, Naresh R.

被引用次数: 174

2018

An In-Memory VLSI Architecture for Convolutional Neural Networks

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Kang, Mingu; Lim, Sungmin; Gonugondla, Sujan; Shanbhag, Naresh R.

被引用次数: 56

2018

Slicer Architectures for Analog-to-Information Conversion in Channel Equalizers

IEEE TRANSACTIONS ON COMMUNICATIONS

Wadhwa, Aseem; Madhow, Upamanyu; Shanbhag, Naresh R.

被引用次数: 6

2017

An Energy-Efficient ECG Processor in 45-nm CMOS Using Statistical Error Compensation (vol 48, pg 2882, 2013)

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Abdallah, Rami A.; Shanbhag, Naresh R.

被引用次数: 0

2016

In-Memory Computing Architectures for Sparse Distributed Memory

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Kang, Mingu; Shanbhag, Naresh R.

被引用次数: 22

2016

Embedded Algorithmic Noise-Tolerance for Signal Processing and Machine Learning Systems via Data Path Decomposition

IEEE TRANSACTIONS ON SIGNAL PROCESSING

Zhang, Sai; Shanbhag, Naresh R.

被引用次数: 22

2016

Error Resilient and Energy Efficient MRF Message-Passing-Based Stereo Matching

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kim, Eric P.; Choi, Jungwook; Shanbhag, Naresh R.; Rutenbar, Rob A.

被引用次数: 5

2016

A 3.6-mW 50-MHz PN Code Acquisition Filter via Statistical Error Compensation in 180-nm CMOS

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kim, Eric P.; Baker, Daniel J.; Narayanan, Sriram; Shanbhag, Naresh R.; Jones, Douglas L.

被引用次数: 3

2015

System-Level Optimization of Switched-Capacitor VRM and Core for Sub/Near-Vt Computing

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhang, Sai; Shanbhag, Naresh R.; Krein, Philip T.

被引用次数: 4

2014

Reducing Energy at the Minimum Energy Operating Point Via Statistical Error Compensation

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Abdallah, Rami A.; Shanbhag, Naresh R.

被引用次数: 2

2014

Robust and Energy Efficient Multimedia Systems via Likelihood Processing

IEEE TRANSACTIONS ON MULTIMEDIA
Abdallah, Rami A.; Shanbhag, Naresh R.
被引用次数: 6

2013

An Energy-Efficient ECG Processor in 45-nm CMOS Using Statistical Error Compensation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Abdallah, Rami A.; Shanbhag, Naresh R.

被引用次数: 41

2013

BER-Optimal Analog-to-Digital Converters for Communication Links

IEEE TRANSACTIONS ON SIGNAL PROCESSING

Narasimha, Rajan; Lu, Minwei; Shanbhag, Naresh R.; Singer, Andrew C.

被引用次数: 34

2012

Soft N-Modular Redundancy

IEEE TRANSACTIONS ON COMPUTERS

Kim, Eric P.; Shanbhag, Naresh R.

被引用次数: 56

2012

Computation as Estimation: A General Framework for Robustness and Energy Efficiency in SoCs

IEEE TRANSACTIONS ON SIGNAL PROCESSING

Narayanan, Sriram; Varatkar, Girish Vishnu; Jones, Douglas L.; Shanbhag, Naresh R.

被引用次数: 7

2010

Stochastic Networked Computation

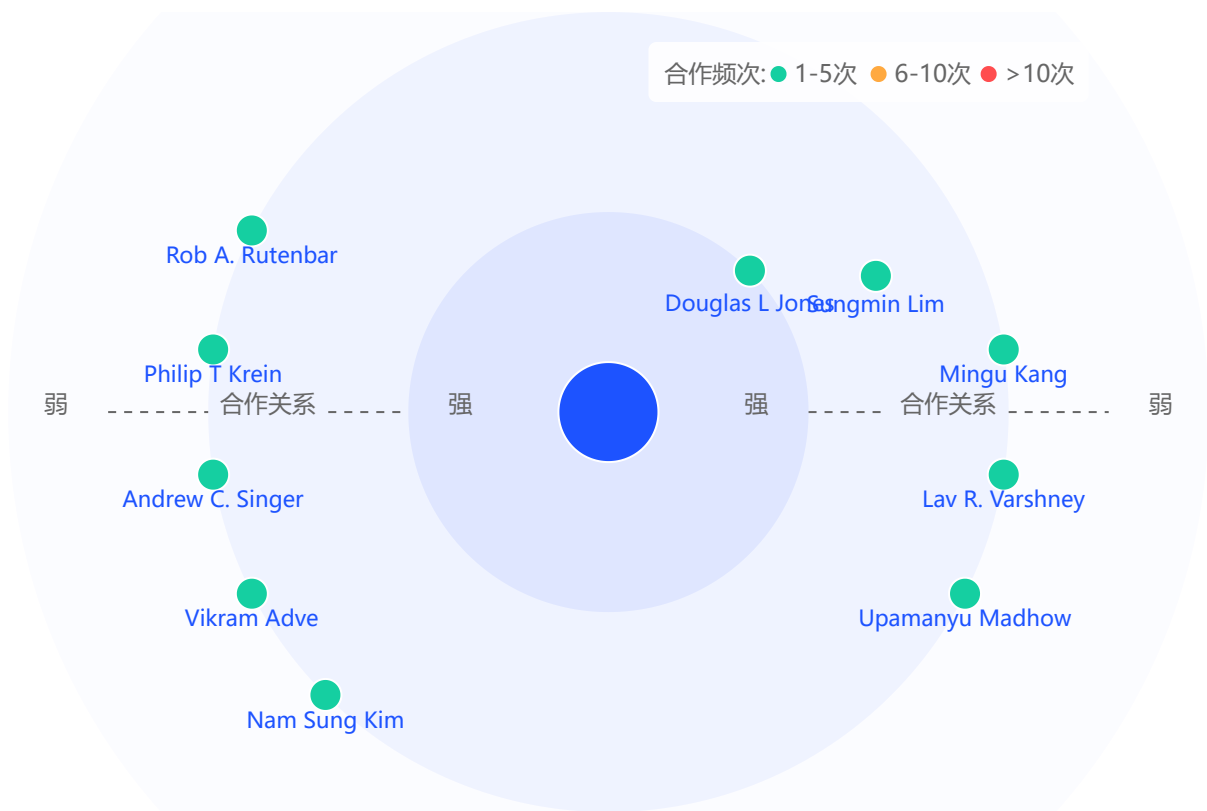
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Varatkar, Girish Vishnu; Narayanan, Sriram; Shanbhag, Naresh R.; Jones, Douglas L.

被引用次数: 14

2010

学术合作



Douglas L Jones

伊利诺伊大学厄巴纳-香槟

合作次数: 3

Sungmin Lim

伊利诺伊大学厄巴纳-香槟

合作次数: 2

Mingu Kang

加州大学-圣地亚哥

合作次数: 1

Lav R. Varshney

伊利诺伊大学厄巴纳-香槟

合作次数: 1

Upamanyu Madhow

加州大学-圣塔芭芭拉

合作次数: 1

还有 5 个合作者